Khyber Pakhtunkhwa Public Service Commission (KPPSC)

Provincial Management Service (PMS) Competitive Examination

GEOGRAPHY — PAPER-I (PHYSICAL AND GENERAL GEOGRAPHY)

Compendium of Solved Past Papers (2008 – 2025)

> **Time Allowed: 3 Hours | Maximum Marks: 100 Marks**
> **Instructions: Attempt ANY FIVE (5) questions. All questions carry equal marks (20 Marks each). Support your answers with relevant geographical sketches, diagrams, and maps where appropriate.**

📝 KPPSC PMS GEOGRAPHY PAPER-I — 2025

- **Q.No.1: Describe the internal structure of the Earth and explain how seismic wave behavior provides evidence for the layering of the Earth's interior. Evaluate the scientific reliability of seismic methods in understanding deep Earth processes.** ***(20 Marks)***
- ***Q.No.2: Explain the origin of ocean tides and critically examine the major theories proposed to explain tidal behavior. How do local coastal configurations influence tidal patterns? (20 Marks)***
- ***Q.No.3: Discuss the major forms of precipitation and the atmospheric mechanisms that produce them. Support your answer with suitable regional and local examples illustrating each type. (20 Marks)***
- ***Q.No.4: Analyze the formation and development of tropical weather systems with reference to hurricanes. Describe the weather patterns associated with hurricanes and the environmental factors influencing their intensity. (20 Marks)***
- ***Q.No.5: Explain the basis of the First Letter Classification in the Köppen Climate Classification System. Compare and contrast the Af (tropical rainforest) and Aw (tropical savanna) climates in terms of distribution, temperature-precipitation regimes, and ecological characteristics. (20 Marks)***
- ***Q.No.6: Discuss the environmental conditions necessary for the development of Karst Landscapes. Describe the major Karst Landforms and evaluate the processes responsible for their evolution. (20 Marks)***
- ***Q.No.7: Explain the principles behind the construction and interpretation of weather maps. How are modern weather forecasts generated, and what are the limitations of current forecasting techniques? (20 Marks)***
- ***Q.No.8: Describe the factors controlling salinity of sea water. Critically analyze the spatial and vertical variations in salinity across the world's oceans and explain their influence on ocean circulation and marine life. (20 Marks)***

📝 *KPPSC PMS GEOGRAPHY PAPER-I — 2022*

- ***Q.No.1: Discuss various processes of physical and chemical weathering with support from sketches/diagrams. Explain their geomorphic significance. (20 Marks)***

- ***Q.No.2: Provide a detailed account of sedimentary rocks and their most workable classification based on origin and composition. (20 Marks)***
- ***Q.No.3: Discuss the distribution of pressure belts over the globe and the resulting planetary wind circulation systems. (20 Marks)***
- ***Q.No.4: Air masses and fronts: Give a comprehensive classification and elaborate their weather characteristics with sketches. (20 Marks)***
- ***Q.No.5: Describe the composition and vertical structure of the atmosphere. Explain the temperature profile of each layer. (20 Marks)***
- ***Q.No.6: Discuss valley glaciers and the resultant erosional and depositional landforms in detail. (20 Marks)***
- ***Q.No.7: Analyze the major causes of salinity variation in the oceans of the world. (20 Marks)***
- ***Q.No.8: Write comprehensive short notes on any TWO of the following: (10 + 10 = 20 Marks)***

(a) Maps and types of maps

(b) Merits and demerits of cylindrical projections

(c) Map projections and their choice in cartography

(d) Ocean currents of the Atlantic Ocean

📝 *KPPSC PMS GEOGRAPHY PAPER-I — 2018*

- ***Q.No.1: Discuss the endogenic and exogenic geomorphic processes in detail. How do they shape the Earth's relief? (20 Marks)***
- ***Q.No.2: What are the major erosional and depositional landforms produced by rivers? Illustrate their evolution from youth to old age. (20 Marks)***
- ***Q.No.3: Discuss the main types of rocks along with examples and describe the geological rock cycle. (20 Marks)***
- ***Q.No.4: Discuss the horizontal distribution of temperature and factors affecting the temperature distribution across the globe. (20 Marks)***
- ***Q.No.5: Describe the different layers and composition of the atmosphere with reference to lapse rate and pause zones. (20 Marks)***
- ***Q.No.6: Explain with the help of maps the major ocean currents of the Indian, Pacific, and Atlantic Oceans. (20 Marks)***
- ***Q.No.7: Define map and discuss its types along with their practical uses in modern geographical studies. (20 Marks)***
- ***Q.No.8: Write short notes on any TWO of the following: (10 + 10 = 20 Marks)***

(a) Glacial landforms

(b) Mercator's Projection

(c) Ocean tides

(d) Climate change and coping mechanisms

📝 *KPPSC PMS GEOGRAPHY PAPER-I — 2016*

- ***Q.No.1: "The heat energy entering the atmosphere and its part reaching the earth surface is balanced by an equal amount of heat going out into outer space." Discuss the global heat balance. (20 Marks)***
- ***Q.No.2: What are the distinguishing characteristics of an air mass? How does a source region influence the type of an air mass that forms over it? Explain the life cycle of a wave cyclone. (20 Marks)***
- ***Q.No.3: What are the different layers of the Earth? Relate the three types of plate collisions associated with seismicity, volcanism, and mountain building. (20 Marks)***
- ***Q.No.4: Describe the nature of groundwater and define the elements of groundwater environment. Review the processes and features associated with Karst topography. (20 Marks)***
- ***Q.No.5: Explain the chemical composition of seawater and the basic physical structure of the ocean. (20 Marks)***
- ***Q.No.6: "The coastal system is the scene of complex tidal fluctuation, winds, waves, and ocean currents." Discuss coastal processes and coastal landforms. (20 Marks)***
- ***Q.No.7: Define cartography and mapping basics: Explain map scale (Representative Fraction) and the principles of map projections. (20 Marks)***
- ***Q.No.8: Write comprehensive notes on any TWO of the following: (10 + 10 = 20 Marks)***

(a) Planetary wind system

(b) The Köppen climatic classification

(c) Wind deposition / Aeolian landforms

(d) Mercator's projection

KPPSC PMS GEOGRAPHY PAPER-I — 2013

- ***Q.No.1: Write an essay on the Radiation Balance of the Earth and Atmosphere. (20 Marks)***
- ***Q.No.2: What are the causes of Atmospheric Circulation? Explain the large-scale wind systems of the globe. (20 Marks)***
- ***Q.No.3: Define rocks and minerals. Discuss the circumstances under which sedimentary rocks are formed and classified. (20 Marks)***
- ***Q.No.4: Discuss the factors promoting mass movements in weathered materials. Explain the various types of mass movements. (20 Marks)***
- ***Q.No.5: Describe the currents of the Pacific Ocean and their climatic significance. (20 Marks)***
- ***Q.No.6: Write a comprehensive note on the salinity of ocean water and factors governing its variations. (20 Marks)***
- ***Q.No.7: What are Map Projections? Give the merits and demerits of Cylindrical Projections. (20 Marks)***

KPPSC PMS GEOGRAPHY PAPER-I — 2010

- ***Q.No.1: Write an essay on the influence of latitude on the annual cycle of insolation. (20 Marks)***
- ***Q.No.2: Define Cyclone and Anticyclone with particular reference to air movement and circulation in Northern and Southern hemispheres. What weather is associated with each? (20 Marks)***

- ***Q.No.3: Differentiate between Valley Glacier and Ice Sheet. Discuss the depositional landforms of Valley Glacier in detail. (20 Marks)***
- ***Q.No.4: What is an earthquake, and how does it arise? Describe the relationship between earthquakes and plate tectonics. (20 Marks)***
- ***Q.No.5: Describe in detail the various movements of ocean water and the factors responsible. (20 Marks)***
- ***Q.No.6: Why does salinity of ocean water vary from place to place? What are the determining factors? (20 Marks)***
- ***Q.No.7: Define Map. What are the essentials of a map? Why do we need projections for drawing a map? (20 Marks)***
- ***Q.No.8: Write notes on any TWO: (a) Karst Topography (b) Ocean floor morphology (c) Temperature inversion (d) Mercator's Projection. (20 Marks)***

📝 *KPPSC PMS GEOGRAPHY PAPER-I — 2008*

- ***Q.No.1: Describe the rock cycle. How can we identify igneous rocks on the basis of mineral contents? Explain types of sedimentary rocks. (20 Marks)***
- ***Q.No.2: Describe the mechanism and types of seismic waves. Define the causes of earthquakes and write an account of earthquakes in Pakistan. (20 Marks)***
- ***Q.No.3: "Solar input energy is equal to Earth's output energy." Explain. Describe the classification of air masses with examples. (20 Marks)***
- ***Q.No.4: Explain the mechanism of oceanic currents that produce circulation in ocean water. (20 Marks)***
- ***Q.No.5: Define the internal structure of the Earth from Asthenosphere to inner core. Give reasons why the core is composed of iron. (20 Marks)***
- ***Q.No.6: How can we identify different types of maps? Characteristics of Cylindrical and Conical Projections. (20 Marks)***
- ***Q.No.7: Short notes on any TWO: (1) Weathering (2) Volcanic activity (3) River landforms (4) Distribution of temperature. (20 Marks)***
